Exercise 8

Control of ventilation

Consider the following model, which describes CO_2 exchange in the alveoli and tissues.

$$V_{lung}\frac{dP_{aCO2}(t)}{dt} = \left(\dot{V}(t) - \dot{V}_D\right)\left(P_{ICO2} - P_{aCO2}(t)\right) + 863QK_{CO2}\left(P_{vCO2}(t) - P_{aCO2}(t)\right) \quad (1)$$

$$V_{tiss}\alpha_{CO2}\frac{dP_{vCO2}(t)}{dt} = QK_{CO2}\left(P_{aCO2}(t) - P_{vCO2}(t)\right) + \dot{M}_{CO2} \quad (2)$$

The previous model (the plant) is controlled through the following equations, which simulate the actions of the central and peripheral chemoreceptors.

$$\tau_p\frac{d\Delta\dot{V}_p(t)}{dt} = -\Delta\dot{V}_p(t) + G_p\left(P_{aCO2}(t - T_p) - \vartheta_p\right) \quad (3)$$

$$\tau_c\frac{d\Delta\dot{V}_c(t)}{dt} = -\Delta\dot{V}_c(t) + G_c\left(P_{aCO2}(t - T_c) - \vartheta_c\right) \quad (4)$$

$$\dot{V}(t) = \dot{V}_0 + \Delta\dot{V}_p(t) + \Delta\dot{V}_c(t)$$

the state variables are

P_{aCO2}: arterial CO_2 pressure = alveolar CO_2 pressure (in mmHg)

P_{vCO2}: venous CO_2 pressure (in mmHg)

$\Delta\dot{V}_p$: effect of the peripheral control on ventilation (l/s)

$\Delta\dot{V}_c$: effect of the central control on ventilation (l/s)

The inputs are:

Q:	cardiac output (l/s)	[typical value 0.083 l/s]
P_{ICO2}:	CO_2 pressure in the inspired air (in mmHg)	[typical value 0 mmHg]
$\dot{M}_{CO2}$:	CO_2 production rate (in l(STPD)/min)	[typical value 0.25 l(STPD)/min]

The parameters are:

V_{lung}: pulmonary volume	2.5 l
$\dot{V}_D$: ventilation of the dead space	0.03 l/s
K_{CO2}: dissociation constant for CO_2 in blood	0.012 l(STPD) (l mmHg)$^{-1}$

V_{tiss}: tissue volume	40 l
a_{CO2}: dissociation constant for CO_2 in tissue	0.001 l(STPD) (l mmHg)$^{-1}$
$\dot{V}_0$: the basal value of ventilation	0.12 l/s
G_p: gain of the peripheral chemoreceptor control	0.02 l/s/mmHg
G_c: gain of the central chemoreceptor control	0.04 l/s/mmHg
□$_p$: time constant of the peripheral chemoreceptor	20 s
□$_c$: time constant of the central chemoreceptor	120 s
T_p: pure delay of the peripheral control	6.1 s
T_c: pure delay of the central control	7.1 s
ϑ_p ϑ_c : set point of the peripheral and central controls	40 mmHg

1) simulate the system in the absence of a controller ($Gp = Gc = 0$) and find the values in the equilibrium condition [you can start using only equations (1) and (2) and add equations (3) and (4) only at point 3];

2) simulate an increase in P_{ICO2} or in $\dot{M}_{CO2}$ without the control, and look at the effect;

3) repeat the same changes as in point 2 but in the presence of the controllers;

4) simulate the case of a patient with heart failure, reducing cardiac output and proportionally increasing the pure delays to observe the instability and the Cheyne Stokes ventilation.

<u>Suggestion</u>: to simulate the presence of pure delays, calculate the number of steps relative to the greater delay, using the variable

start = max([round(Tp/dt) round(Tc/dt)])

where dt is the integration step. We apply Euler's method with an index j > start. Conversely, when j < start, keep pressures and ventilations at baseline and ventilation controls at zero.

1. Model equations

The state variables are **$Paco_2$** (arterial/alveolar CO_2 pressure), **$Pvco_2$** (venous CO_2 pressure), **DVp** (peripheral ventilatory contribution), and **DVc** (central ventilatory contribution). The total ventilation is

$$V(t) = V0 + DVp(t) + DVc(t)$$

The plant is described by the following differential equations:

$$dPaco2/dt = [\ (V - VD)\cdot(PIco2 - Paco2) + 863\cdot Q\cdot Kco2\cdot(Pvco2 - Paco2)\ ] / Vlung$$

$$dPvco2/dt = [\ Q\cdot Kco2\cdot(Paco2 - Pvco2) + Mco2\ ] / (Vtiss\cdot alphaCO2)$$

The two controllers are first-order delayed chemoreflexes:

$$dDVp/dt = [\ -DVp + Gp\cdot(Paco2(t - Tp) - thetap)\ ] / taup$$

$$dDVc/dt = [\ -DVc + Gc\cdot(Paco2(t - Tc) - thetac)\ ] / tauc$$

Using the parameters given in the exercise, the baseline values are: Vlung = 2.5 L, VD = 0.03 L/s, V0 = 0.12 L/s, Q = 0.083 L/s, Mco_2 = 0.25/60 L(STPD)/s, Kco_2 = 0.012 L(STPD)/(L·mmHg), Vtiss = 40 L, $alphaCO_2$ = 0.001 L(STPD)/(L·mmHg).

2. Equilibrium without controller

For point 1, the controller is absent, so **Gp = Gc = 0**, hence **DVp = DVc = 0** and the ventilation remains at the basal value **V = V0 = 0.12 L/s** (that is, **7.2 L/min**). At equilibrium, by imposing **$dPaco_2/dt = 0$** and **$dPvco_2/dt = 0$**, one obtains:

$$Paco2,eq = 863\cdot Mco2 / (V0 - VD) \approx 39.95 \text{ mmHg}$$

$$Pvco2,eq = Paco2,eq + Mco2 / (Q\cdot Kco2) \approx 44.14 \text{ mmHg}$$

Therefore, the uncontrolled system starts near the physiological operating point: **$Paco_2$ ≈ 40 mmHg** and **$Pvco_2$ ≈ 44 mmHg**. These are the correct equilibrium values to initialize the simulation.

3. Increase in metabolism or inspired CO_2 without control

If **Mco_2** increases while ventilation stays fixed, more CO_2 is produced by the tissues than the lungs can eliminate. As a consequence, both **$Paco_2$** and **$Pvco_2$** rise until a new equilibrium is reached. The increase in **$Pvco_2$** is usually larger because venous blood is closer to the production site.

If **$PIco_2$** increases, the inspired air already contains CO_2, so the alveolar compartment cannot wash out carbon dioxide effectively. **$Paco_2$** increases directly and **$Pvco_2$** increases secondarily. In both cases, **without control**, ventilation does not adapt, so hypercapnia persists.

4. Effect of peripheral and central chemoreceptors

When the feedback is activated, the model becomes: **V = V0 + DVp + DVc**. If **$Paco_2$** rises above the 40 mmHg set point, both controllers increase ventilation. The **peripheral control** is faster (**taup = 20 s**) but weaker (**Gp = 0.02 L/s/mmHg**), whereas the **central control** is slower (**tauc = 120 s**) but stronger (**Gc = 0.04 L/s/mmHg**).

As a result, after an increase in metabolism or inspired CO_2, the controlled system shows:

- a smaller rise in $Paco_2$ and $Pvco_2$ compared with the uncontrolled case;
- an increase in ventilation above the basal value V0;
- a fast initial correction due to the peripheral controller and a slower sustained correction due to the central controller.

In the MATLAB implementation, this requires uncommenting the equations for **DDVp** and **DDVc**, and using delayed samples **Paco2(j - round(Tp/dt))** and **Paco2(j - round(Tc/dt))**, exactly as suggested in the exercise text.

5. Heart failure and Cheyne–Stokes ventilation

In heart failure, cardiac output **Q** is reduced and the circulation time increases. In the model, this is represented by lowering **Q** and proportionally increasing the pure delays **Tp** and **Tc**.

This has two important consequences:

- the transport of CO_2 from tissues to lungs is slower;
- the controller reacts to an old value of $Paco_2$ rather than to the current one.

The feedback therefore becomes poorly damped. Instead of converging smoothly, ventilation may overshoot and undershoot cyclically, producing oscillations in **$Paco_2$** and in total ventilation.

This is the mechanism behind **Cheyne–Stokes breathing**: alternating phases of hyperventilation and hypoventilation/apnea caused by excessive delay in the control loop.

6. Conclusion

This exercise shows that the respiratory control system is a delayed feedback loop. **Without control**, increases in metabolism or inspired CO_2 lead to sustained hypercapnia. **With peripheral and central chemoreceptors**, ventilation rises and partially restores normal CO_2 pressure. However, when circulation is slowed and delays become long, the closed-loop system can become oscillatory and generate periodic breathing patterns such as Cheyne–Stokes ventilation.